2026-2027 I SEMESTER MTL 712/MTL 7012
Lab Sheet-I

Instructions

1. Implement all numerical methods from scratch. Do **not** use any built-in ODE solvers.
2. You may use built-in functions only for basic arithmetic, plotting, and computing the exact solution for comparison.
3. Include plots with appropriate axis labels, legends, and titles.

1 Euler's Method

Use Euler's method with step size $h = 0.1$ to approximate the solution of the IVP

$$y' = y - x^2 + 1, \qquad y(0) = 0.5, \qquad 0 \leq x \leq 1.$$

Compare the numerical solution with the exact solution

$$y(x) = (x+1)^2 - \frac{1}{2}e^x.$$

Discuss how the error grows as x increases. Also, plot the numerical solution obtained using Euler's method together with the exact solution on the same graph.

2 Backward Euler's Method

Apply the backward Euler method with $h = 0.1$ to solve the IVP

$$y' = -50y, \qquad y(0) = 1, \qquad 0 \leq x \leq 1.$$

Since the right-hand side is linear in y, derive an explicit formula for y_{n+1} in terms of y_n. Compare the numerical solution with the exact solution $y(x) = e^{-50x}$, discuss the stability of the method in comparison with the (forward) Euler method, and plot the numerical and exact solutions on the same graph.

3 Improved Euler's Method

Use the improved Euler method (Heun's method) with $h = 0.1$ to solve the IVP

$$y' = x + y, \qquad y(0) = 1, \qquad 0 \le x \le 1.$$

Compare the numerical solution with the exact solution

$$y(x) = 2e^x - x - 1.$$

Plot the numerical solution obtained using the improved Euler method together with the exact solution on the same graph. Comment on the accuracy improvement achieved by the corrector step over the simple Euler method.

4 Modified Euler's Method

Use the modified Euler method (midpoint method) with $h = 0.1$ to solve the IVP

$$y' = xy, \qquad y(0) = 1, \qquad 0 \le x \le 1.$$

Compare the numerical solution with the exact solution

$$y(x) = e^{x^2/2}.$$

Repeat the computation with $h = 0.05$ and plot the exact solution together with the numerical solutions obtained using both step sizes on the same graph. Discuss the order of accuracy observed as the step size is halved.

5 Classical Fourth-Order Runge–Kutta Method

Apply the classical fourth-order Runge–Kutta method with $h = 0.2$ to solve the IVP

$$y' = \cos(x) - y, \qquad y(0) = 1, \qquad 0 \le x \le 2.$$

Compare the numerical solution with the exact solution

$$y(x) = \frac{1}{2}(\sin x + \cos x) + \frac{1}{2}e^{-x}.$$

Plot the numerical and exact solutions on the same graph, and comment on the accuracy achieved relative to the lower-order methods used earlier in this lab.

6 Comparative Study

Solve the IVP

$$y' = y(1 - y), \qquad y(0) = 0,5, \qquad 0 \le x \le 1$$

using all five numerical methods studied in this lab with a uniform step size of $h = 0.25$ (and $h = 0.125$).

Prepare tables showing the mesh points x_n, the numerical approximations obtained using each method. Plot the numerical solutions obtained from all the five methods on the same graph for each step size.